

NIVELL 1

EXERCICI 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.

```
6 • DROP DATABASE transactions;
7 • DROP TABLE IF EXISTS company;
8 • DROP TABLE IF EXISTS transaction;
9
10 # 1) Executar aquí els fitxers "estructura_dades" i "dades_introduir".
11 # 2) Comprovar l'estructura creada:
12
13 • USE transactions;
14 • SHOW TABLES;
15 • DESCRIBE company;
16 • DESCRIBE transaction;
17
18 # Executed 9 (16114ms) 30% completed for database transactions.
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents: |

Tables_in_transactions

- company
- credit_card
- transaction

Result 47 x | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	17:37:09	USE transactions	0 row(s) affected	0.000 sec
2	17:37:13	SHOW TABLES	3 row(s) returned	0.000 sec / 0.000 sec

```
13 • USE transactions;
14 • SHOW TABLES;
15 • DESCRIBE company;
16 • DESCRIBE transaction;
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents: |

Field	Type	Null	Key	Default	Extra
id	varchar(15)	NO	PRI		
company_name	varchar(255)	YES			
phone	varchar(15)	YES			
email	varchar(100)	YES			
country	varchar(100)	YES			
website	varchar(255)	YES			

Result 48 x | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	17:37:13	SHOW TABLES	3 row(s) returned	0.000 sec / 0.000 sec
3	17:39:39	DESCRIBE company	6 row(s) returned	0.000 sec / 0.000 sec

```
13 • USE transactions;
14 • SHOW TABLES;
15 • DESCRIBE company;
16 • DESCRIBE transaction;
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents: |

Field	Type	Null	Key	Default	Extra
id	varchar(255)	NO	PRI		
credit_card_id	varchar(15)	YES	MUL		
company_id	varchar(20)	YES	MUL		
user_id	int	YES			
lat	float	YES			
longitude	float	YES			
timestamp	timestamp	YES			
amount	decimal(10,2)	YES			
dedined	tinyint(1)	YES			

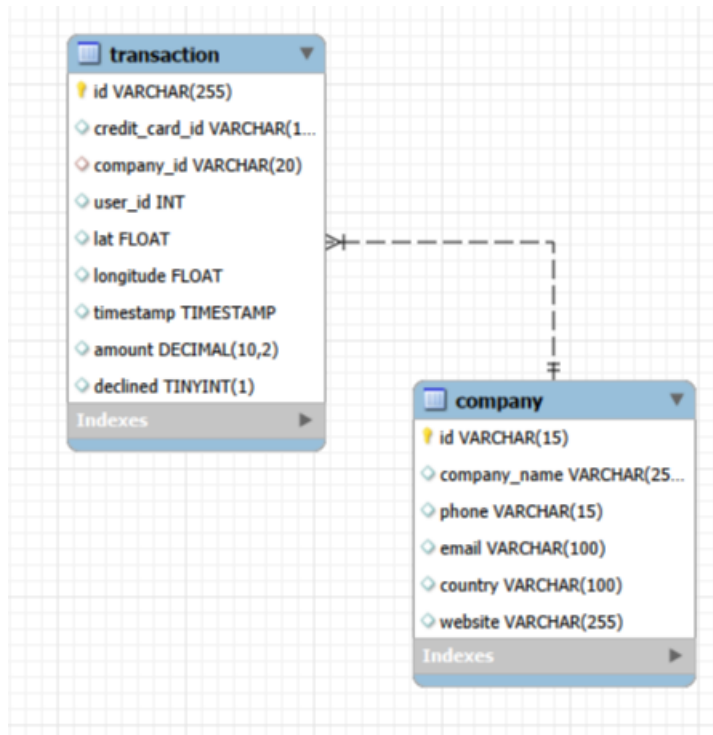
Result 49 x | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	17:39:39	DESCRIBE company	6 row(s) returned	0.000 sec / 0.000 sec
4	17:44:58	DESCRIBE transaction	9 row(s) returned	0.000 sec / 0.000 sec

DIAGRAMA:



Troben 2 taules:

1. **company**: columnes (**PK id**, company_name, phone, email, country, website)
2. **transaction**: columnes (**PK id**, **FK company_id**, credit_card_id, user_id, lat, longitude, timestamp, amount, declined)

La relació de cardinalitat és de **1:N** (una botiga pot realitzar moltes vendes però cada venda pertany a una única botiga).

La columna **company_id** de la taula **transaction** actua com a clau forana i es relaciona amb la columna **id** de la taula **company**.

La taula **company** representa les botigues. Cada fila correspon a una botiga diferent i cada columna emmagatzema una característica d'aquesta botiga, com ara el nom, el telèfon, el país o el lloc web.

La taula **transaction** representa les vendes realitzades. Cada fila correspon a una venda i està associada a una única empresa mitjançant el camp **company_id**.

EXERCICI 2.1: Llistat dels països que estan generant vendes

```
18 # Exercici 2. Utilitzant JOIN realitzaràs les següents consultes:
19 # 2.1 Llistat dels països que estan generant vendes.
20
21 • SELECT DISTINCT country
22 FROM company c
23 JOIN `transaction` t ON c.id = t.company_id
24 WHERE "declined" = 0;
25
```

Result Grid

country
Germany
Australia
United States
New Zealand
Norway

Result 58 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:11:10	SELECT DISTINCT country FROM company c JOIN `transaction` t ON c.id = t.company_id WHERE "declined" = 0;	16 row(s) returned	0.015 sec / 0.000 sec

EXERCICI 2.2: Des de quants països es generen les vendes?

```
26 # 2.2 Des de quants països es generen les vendes.
27 • SELECT COUNT(DISTINCT country) AS total_paisos
28 FROM company c
29 JOIN `transaction` t ON c.id = t.company_id
30 WHERE "declined" = 0;
31
```

Result Grid

total_paisos
15

Result 60 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	10:18:43	SELECT COUNT(DISTINCT country) AS total_paisos FROM company c JOIN `transaction` t ON c.id = t.co...	1 row(s) returned	0.141 sec / 0.000 sec
3	10:21:30	SELECT COUNT(DISTINCT country) AS total_paisos FROM company c JOIN `transaction` t ON c.id = t.co...	1 row(s) returned	0.125 sec / 0.000 sec

EXERCICI 2.3: Identifica la companyia amb la mitjana més gran de vendes

```
32 # 2.3 Identifica la companyia amb la mitjana més gran de vendes.
33 • SELECT mitjana_vendes_companyia.id, mitjana_vendes_companyia.company_name, max_mitjana_vendes.max_mitjana
34 FROM (
35     SELECT c.id, c.company_name, ROUND(AVG(t.amount),2) AS mitjana_vendes
36     FROM company c
37     JOIN `transaction` t ON c.id = t.company_id
38     WHERE "declined" = 0
39     GROUP BY c.id) AS mitjana_vendes_companyia
40 JOIN (
41     SELECT MAX(mitjana_vendes) AS max_mitjana
42     FROM (
43         SELECT c.id, c.company_name, ROUND(AVG(t.amount),2) AS mitjana_vendes
44         FROM company c
45         JOIN `transaction` t ON c.id = t.company_id
46         WHERE "declined" = 0
47         GROUP BY c.id) AS mitjana_vendes_companyia2
48     ) AS max_mitjana_vendes
49 ON mitjana_vendes_companyia.mitjana_vendes = max_mitjana_vendes.max_mitjana;
```

Result Grid

id	company_name	max_mitjana
b-2222	Ac Fermentum Incorporated	284.87

Result 61 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	10:21:30	SELECT COUNT(DISTINCT country) AS total_paisos FROM company c JOIN `transaction` t ON c.id = t.co...	1 row(s) returned	0.125 sec / 0.000 sec
4	10:28:25	SELECT mitjana_vendes_companyia.id, mitjana_vendes_companyia.company_name, max_mitjana_vendes...	1 row(s) returned	1.516 sec / 0.000 sec

EXERCICI 3. Utilitzant només subconsultes (sense utilitzar JOIN)

```
52 # 3.1 Mostra totes les transaccions realitzades per empreses d'Alemanya. --> S'agafa com a realitzades totes les transaccions (completades i rebutjades "declined").
53 SELECT *
54 FROM `transaction` t
55 WHERE EXISTS (
56     SELECT *
57     FROM company c
58     WHERE c.id = t.company_id
59     AND c.country = 'Germany'
60 )
61
```

Result Grid

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
0013303B-206D-4C03-94B7-63A2676EB984	Cc5-4899	b-2222	318	41.3781	12.447	2020-03-25 10:43:43	426.36	0
0013C1B6-3894-406C-8154-E2B3FECABE9	Cc5-5070	b-2222	489	41.3814	2.18176	2020-12-17 18:15:37	316.90	0
00201A11-2E62-44C4-941D-198FC8DB77F0	CcJ-3512	b-2222	193	55.5704	-3.65129	2021-01-22 23:44:27	453.04	0
00235618-0A5C-4D49-9DCB-83A9405D8923	Cc5-8137	b-2222	3556	59.8421	18.729	2020-09-09 15:43:19	263.14	0
005A5A78-1F1A-4B6C-9B15-1625A78C9C38	Cc5-8998	b-2222	4417	41.1591	-8.63905	2024-05-15 09:10:11	442.01	0

transaction 62 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	10:28:25	SELECT mitjana_vendes_companya.id, mitjana_vendes_companya.company_name, max_mitjana_vendes...	1 row(s) returned	1.516 sec / 0.000 sec
5	10:38:25	SELECT * FROM `transaction` t WHERE EXISTS (SELECT * FROM company c WHERE c.id = t.company...	13291 row(s) returned	0.000 sec / 0.078 sec

```
62 # 3.2 Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.
63 SELECT id, company_name
64 FROM company c
65 WHERE EXISTS (
66     SELECT *
67     FROM `transaction` t
68     WHERE t.company_id = c.id
69     AND t.amount > (
70         SELECT AVG(amount) AS mitjana_transaccions
71         FROM `transaction` t
72     )
73 )
74
```

Result Grid

id	company_name
b-2222	Ac Fermentum Incorporated
b-2226	Magna A Neque Industries
b-2230	Fusce Corp.
b-2234	Convallis In Incorporated
b-2238	Ante Iaculis Nec Foundation

company 64 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	10:48:16	SELECT id, company_name FROM company c WHERE EXISTS (SELECT * FROM `transaction` t ...	100 row(s) returned	0.172 sec / 0.000 sec

```
74 # 3.3 Eliminaren del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.
75 SELECT *
76 FROM company c
77 WHERE NOT EXISTS (
78     SELECT company_id
79     FROM `transaction` t
80     WHERE c.id = t.company_id
81 )
82
```

Result Grid

id	company_name	phone	email	country	website
*	id	id	id	id	id

company 65 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	10:49:43	SELECT id, company_name FROM company c WHERE EXISTS (SELECT * FROM `transaction` t ...	100 row(s) returned	0.172 sec / 0.000 sec
8	10:53:31	SELECT * FROM company c WHERE NOT EXISTS (SELECT company_id FROM `transaction` t ...	0 row(s) returned	0.000 sec / 0.000 sec

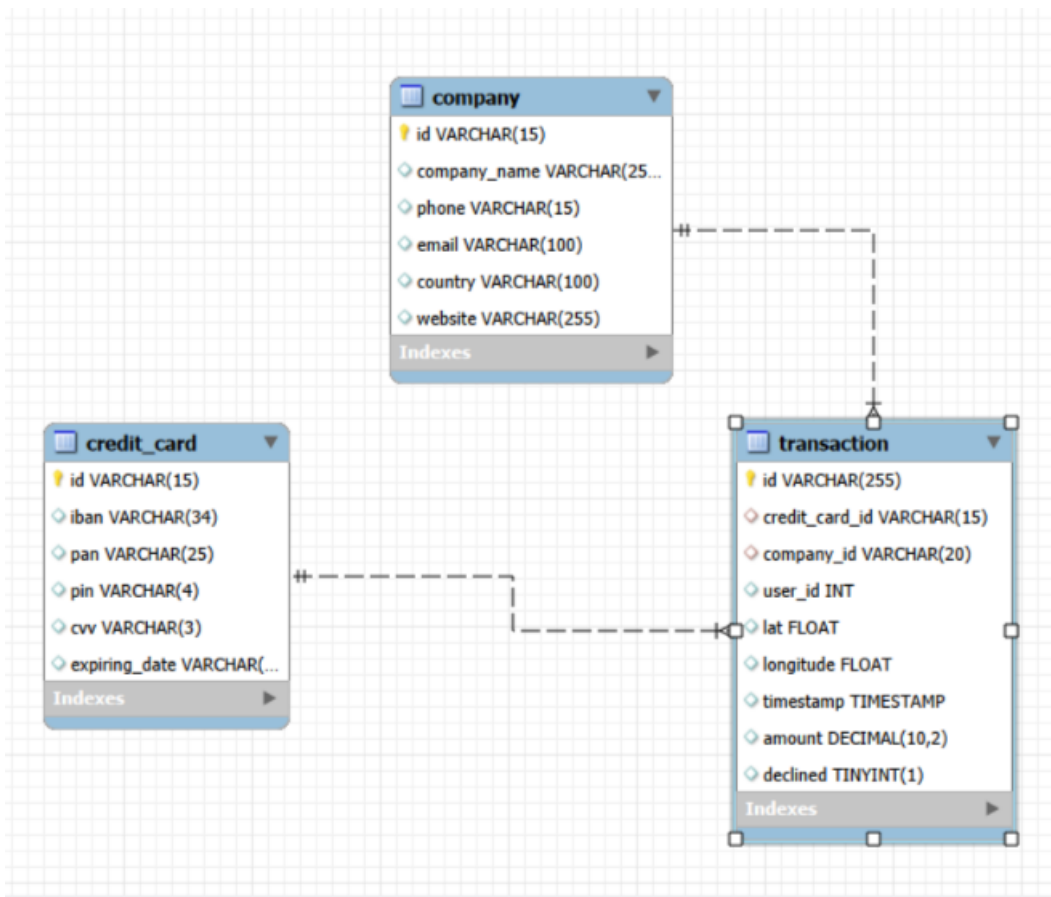
EXERCICI 4

La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.

```
79 • DROP TABLE IF EXISTS credit_card;
80
81 • CREATE TABLE credit_card (
82     id varchar(100) PRIMARY KEY,
83     iban varchar(34),
84     pan varchar(25),
85     pin varchar(4),
86     cvv varchar(3),
87     expiring_date varchar(8)
88 );
89
90 • ALTER TABLE credit_card
91     MODIFY COLUMN id VARCHAR(15);
92
93 • ALTER TABLE transaction
94     ADD CONSTRAINT fk_transaction_credit_card
95     FOREIGN KEY (credit_card_id)
96     REFERENCES credit_card(id);
```

Action Output			
#	Time	Action	Message
19	13:07:04	ALTER TABLE credit_card MODIFY COLUMN id VARCHAR(15)	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0
20	13:09:34	ALTER TABLE transaction ADD CONSTRAINT fk_transaction_credit_card FOREIGN KEY (credit_card_id)...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0
			Duration / Fetch
			0.094 sec
			2.500 sec

Per a poder treballar amb el diagrama de MYSQL Workbench, calia modificar el nombre de varchar de les key relacionades, per aquest motiu s'ha aplicat codi per modificar la varchar i per assignar la correspondència de FK que no s'havia realitzat prèviament.



El diagrama mostra 3 taules (**company**, **credit_card** i **transaction**).

La columna `company_id` de la taula **transaction** actua com a FK i es relaciona amb la columna `id` (PK) de la taula **company**. La relació de cardinalitat és de **1:N** (una companyia pot realitzar moltes transaccions però cada transacció pertany a una única companyia).

La columna `id` de la taula **credit_card** actúa com a PK i es relaciona amb la taula **transaction** (FK `credit_card_id`).

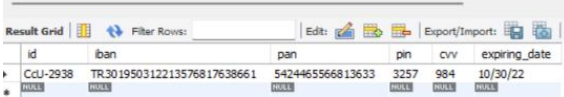
Cada fila de la taula **credit_card** representa una targeta, la qual pot realitzar diverses transaccions **1:N**.

EXERCICI 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.

1) Mostra dades actuals:

```
82 # Exercici 5: El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938.
83 # La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.
84
85 • select *
86   from credit_card
87  WHERE id = "CcU-2938";
88
89 • UPDATE credit_card
90   SET iban = "TR323456312213576817699999"
91  WHERE id = "CcU-2938";
```



id	iban	pan	pin	cvv	expiring_date
CcU-2938	TR301950312213576817638661	5424465566813633	3257	984	10/30/22

credit_card 62 x

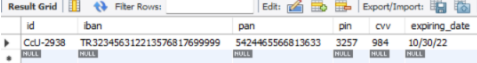
Output

Action Output

#	Time	Action	Message
1	13.46:10	select * from credit_card WHERE id = "CcU-2938"	1 row(s) returned

2) Modifica dades i mostra-les actualitzades:

```
89 • UPDATE credit_card
90   SET iban = "TR323456312213576817699999"
91  WHERE id = "CcU-2938";
92
93 • select *
94   from credit_card
95  WHERE id = "CcU-2938";
```



id	iban	pan	pin	cvv	expiring_date
CcU-2938	TR323456312213576817699999	5424465566813633	3257	984	10/30/22

credit_card 63 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	13.47:02	UPDATE credit_card SET iban = "TR323456312213576817699999" WHERE id = "CcU-2938"	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
3	13.47:28	select * from credit_card WHERE id = "CcU-2938"	1 row(s) returned	0.000 sec / 0.000 sec

Exercici 6

En la taula "transaction" ingressa una nova transacció amb la següent informació:

```
Id
10881D1D-5B23-A76C-55EF-C568E49A99D0

credit_card_id

CcU-9999

company_id

b-9999

user_id

9999

lat

829.999

longitude

-117.999

amount

111.11

declined

0
```

```
97 # Exercici 6. En la taula "transaction" ingressa una nova transacció amb la següent informació:
98 • INSERT INTO company (id)
99   VALUES ("b-9999");
100
101 • INSERT INTO credit_card (id)
102   VALUES ("CcU-9999");
103
104 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)
105   VALUES ("10881D1D-5B23-A76C-55EF-C568E49A99D0", "CcU-9999", "b-9999", "9999", "829.999", "-117.999", "111.11", "0");
106
107 • SELECT *
108   FROM transaction
```

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
10881D1D-5B23-A76C-55EF-C568E49A99D0	CcU-9999	b-9999	9999	829.999	-117.999	111.11	0	

transaction 67 x

Output

#	Time	Action	Message	Duration / Fetch
5	10:38:26	INSERT INTO credit_card (id) VALUES ("CcU-9999")	1 row(s) affected	0.016 sec
6	10:38:37	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUE...	1 row(s) affected	0.016 sec
7	10:39:03	SELECT * FROM transaction	100001 row(s) returned	0.000 sec / 0.454 sec
8	10:39:44	SELECT * FROM transaction WHERE id = "10881D1D-5B23-A76C-55EF-C568E49A99D0"	1 row(s) returned	0.000 sec / 0.000 sec

EXERCICI 7

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card.
Recorda mostrar el canvi realitzat.

1) Taula actual:

```
111 # Exercici 7. Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recorda mostrar el canvi realitzat.
112
113 • SELECT *
114 FROM credit_card;
```

Result Grid

	id	iban	pan	pin	cvv	expiring_date
▶	CcS-4857	XX4857591835292505850771	2314242385113924	1819	467	09/27/25
	CcS-4858	XX8581768137002436094025	6582720299715533	3964	817	12/28/28
	CcS-4859	XX7826930491423553609370	8861684536289642	4983	277	11/26/26
	CcS-4860	XX5559590368835304645299	2481155515498459	6876	661	07/27/27
	CcS-4861	XX2035182877195191627307	1308930301149557	5710	398	04/25/26
	CcS-4862	XX4774721462463645409758	6715617009807829	4042	174	11/27/26
	CcS-4863	XX1476879664745046707111	7140879810451504	5969	440	17/77/79

credit_card 75 x

Output

Action Output

#	Time	Action	Message
1	10:45:12	SELECT * FROM credit_card	5001 row(s) returned

2) Taula actualitzada:

```
115
116 • ALTER TABLE credit_card
117 DROP COLUMN pan;
118
119 • SELECT *
120 FROM credit_card;
```

Result Grid

	id	iban	pin	cvv	expiring_date
▶	CcS-4857	XX4857591835292505850771	1819	467	09/27/25
	CcS-4858	XX8581768137002436094025	3964	817	12/28/28
	CcS-4859	XX7826930491423553609370	4983	277	11/26/26
	CcS-4860	XX5559590368835304645299	6876	661	07/27/27
	CcS-4861	XX2035182877195191627307	5710	398	04/25/26
	CcS-4862	XX4774721462463645409758	4042	174	11/27/26
	CcS-4863	XX1476879664745046707111	5969	440	17/77/79

credit_card 76 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:45:12	SELECT * FROM credit_card	5001 row(s) returned	0.000 sec / 0.000 sec
2	10:50:54	ALTER TABLE credit_card DROP COLUMN pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
3	10:51:08	SELECT * FROM credit_card	5001 row(s) returned	0.000 sec / 0.015 sec

EXERCICI 8

Descarrega els arxius CSV que trobaràs a l'apartat de recursos: american_users.csv, european_users.csv, companies.csv, credit_cards.csv, transactions.csv

Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes.

1) Crear base de dades i carregar taules:

```
149 • DROP DATABASE IF EXISTS nova_bd_transactions;
150 • CREATE DATABASE nova_bd_transactions;
151 • USE nova_bd_transactions;
152
153 • DROP TABLE IF EXISTS companies;
154
155 • CREATE TABLE companies (
156     company_id VARCHAR(255) PRIMARY KEY,
157     company_name VARCHAR(255) NULL,
158     phone VARCHAR(255) NULL,
159     email VARCHAR(255) NULL,
160     country VARCHAR(255) NULL,
161     website VARCHAR(255) NULL,
162     merchant_category VARCHAR(255) NULL,
163     merchant_price_position VARCHAR(255) NULL
164 );
165
166 • DROP TABLE IF EXISTS credit_cards;
```

Output

#	Time	Action	Message	Duration / Fetch
2	11:26:41	CREATE DATABASE nova_bd_transactions	1 row(s) affected	0.000 sec
3	11:26:45	USE nova_bd_transactions	0 row(s) affected	0.000 sec
4	11:26:50	DROP TABLE IF EXISTS companies	0 row(s) affected. 1 warning(s): 1051 Unknown table 'nova_bd_transactions.companies'	0.000 sec
5	11:29:00	CREATE TABLE companies (company_id VARCHAR(255) PRIMARY KEY, company_name VARCHAR...	0 row(s) affected	0.015 sec

```
166 • DROP TABLE IF EXISTS credit_cards;
167
168 • CREATE TABLE credit_cards (
169     id VARCHAR(255) PRIMARY KEY,
170     user_id VARCHAR(255) NULL,
171     iban VARCHAR(255) NULL,
172     pan VARCHAR(255) NULL,
173     pin VARCHAR(255) NULL,
174     cvv VARCHAR(255) NULL,
175     track1 VARCHAR(255) NULL,
176     track2 VARCHAR(255) NULL,
177     expiring_date VARCHAR(255) NULL,
178     card_type VARCHAR(255) NULL,
179     card_renewal_flag VARCHAR(255) NULL
180 );
181
182 • DROP TABLE IF EXISTS transactions;
```

Output

#	Time	Action	Message	Duration / Fetch
5	11:29:00	CREATE TABLE companies (company_id VARCHAR(255) PRIMARY KEY, company_name VARCHAR...	0 row(s) affected	0.015 sec
6	11:30:16	DROP TABLE IF EXISTS credit_cards	0 row(s) affected. 1 warning(s): 1051 Unknown table 'nova_bd_transactions.credit_cards'	0.000 sec
7	11:30:36	CREATE TABLE credit_cards (id VARCHAR(255) PRIMARY KEY, user_id VARCHAR(255) NULL, ib...	0 row(s) affected	0.015 sec

```
182 • DROP TABLE IF EXISTS transactions;
183
184 • CREATE TABLE transactions (
185     id VARCHAR(255) PRIMARY KEY,
186     card_id VARCHAR(255) NULL,
187     business_id VARCHAR(255) NULL,
188     timestamp VARCHAR(255) NULL,
189     amount VARCHAR(255) NULL,
190     declined VARCHAR(255) NULL,
191     product_ids VARCHAR(255) NULL,
192     user_id VARCHAR(255) NULL,
193     lat VARCHAR(255) NULL,
194     longitude VARCHAR(255) NULL,
195     discount_amount VARCHAR(255) NULL,
196     tax_amount VARCHAR(255) NULL,
197     shipping_amount VARCHAR(255) NULL,
198     channel VARCHAR(255) NULL,
199     campaign_id VARCHAR(255) NULL,
200     device_type VARCHAR(255) NULL,
201     is_international VARCHAR(255) NULL,
202     decline_reason VARCHAR(255) NULL,
203     distance_km VARCHAR(255) NULL
204 );
```

Output

#	Time	Action	Message	Duration / Fetch
8	11:30:47	DROP TABLE IF EXISTS transactions	0 row(s) affected. 1 warning(s): 1051 Unknown table 'nova_bd_transactions.transactions'	0.000 sec
9	11:32:25	CREATE TABLE transactions (id VARCHAR(255) PRIMARY KEY, card_id VARCHAR(255) NULL, b...	0 row(s) affected	0.015 sec

```
208 CREATE TABLE users (  
209     id VARCHAR(255) PRIMARY KEY,  
210     name VARCHAR(255) NULL,  
211     surname VARCHAR(255) NULL,  
212     phone VARCHAR(255) NULL,  
213     email VARCHAR(255) NULL,  
214     birth_date VARCHAR(255) NULL,  
215     country VARCHAR(255) NULL,  
216     city VARCHAR(255) NULL,  
217     postal_code VARCHAR(255) NULL,  
218     address VARCHAR(255) NULL,  
219     signup_date VARCHAR(255) NULL,  
220     user_segment VARCHAR(255) NULL,  
221     income_band VARCHAR(255) NULL  
222 );  
223  
224 LOAD DATA  
225 INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_companies.csv"  
226 INTO TABLE companies  
  
Output  
# Time Action Message Duration / Fetch  
10 11:33:41 DROP TABLE IF EXISTS users 0 row(s) affected. 1 warning(s): 1051 Unknown table 'hova_bd_transactions.users' 0.015 sec  
11 11:33:48 CREATE TABLE users ( id VARCHAR(255) PRIMARY KEY, name VARCHAR(255) NULL, surname ... 0 row(s) affected 0.032 sec  
  
224 LOAD DATA  
225 INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_companies.csv"  
226 INTO TABLE companies  
227 FIELDS TERMINATED BY ','  
228 ENCLOSED BY ''''  
229 IGNORE 1 ROWS;  
230  
231 LOAD DATA  
232 INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_credit_cards.csv"  
233 INTO TABLE credit_cards  
234 FIELDS TERMINATED BY ','  
235 ENCLOSED BY ''''  
236 IGNORE 1 ROWS;  
237  
238 LOAD DATA  
239 INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_transactions.csv"  
240 INTO TABLE transactions  
241 FIELDS TERMINATED BY ';'   
242 ENCLOSED BY ''''  
243 IGNORE 1 ROWS;  
244  
Output  
# Time Action Message Duration / Fetch  
24 11:15:32 CREATE TABLE users ( id VARCHAR(255) PRIMARY KEY, name VARCHAR(255) NULL, surname... 0 row(s) affected 0.062 sec  
25 11:17:19 LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_companies.csv" IN... 100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0 0.016 sec  
26 11:17:26 LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_credit_cards.csv" I... 5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0 0.172 sec  
27 11:17:44 LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_transactions.csv" I... 100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0 4.219 sec  
  
245 LOAD DATA  
246 INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_european_users.csv"  
247 INTO TABLE users  
248 FIELDS TERMINATED BY ','  
249 ENCLOSED BY ''''  
250 IGNORE 1 ROWS;  
251  
252 LOAD DATA  
253 INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_american_users.csv"  
254 INTO TABLE users  
255 FIELDS TERMINATED BY ','  
256 ENCLOSED BY ''''  
257 IGNORE 1 ROWS;  
258  
Output  
# Time Action Message Duration / Fetch  
15 11:36:00 LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_european_users.csv... 3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0 0.125 sec  
16 11:36:03 LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8_american_users.csv"... 1010 row(s) affected Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0 0.016 sec
```

Comprovar que el valor existeix a la taula de fets per assignar FK:

```
306 SELECT transactions.user_id  
307 FROM transactions  
308 LEFT JOIN users  
309 ON transactions.user_id = users.id  
310 WHERE users.id IS NULL;  
  
311  
312 SELECT transactions.card_id  
313 FROM transactions  
314 LEFT JOIN credit_cards  
315 ON transactions.card_id = credit_cards.id  
316 WHERE credit_cards.id IS NULL;  
317  
318  
319 SELECT transactions.business_id  
320 FROM transactions  
321 LEFT JOIN companies  
322 ON transactions.business_id = companies.company_id  
323 WHERE companies.company_id IS NULL;  
324
```

Result Grid | Filter Rows | Export | Wrap Cell Contents: IS

user_id

Result 24 x

Action Output

Time Action Message

1 10:41:17 SELECT transactions.user_id FROM transactions LEFT JOIN users ON transactions.user_id = users.id WHERE... 0 row(s) returned

Result Grid | Filter Rows | Export | Wrap Cell Contents: IS

card_id

Result 26 x

Action Output

Time Action Message

1 10:45:30 SELECT transactions.card_id FROM transactions LEFT JOIN credit_cards ON transactions.card_id = credit_... 0 row(s) returned

Result Grid | Filter Rows | Export | Wrap Cell Contents: IS

business_id

Result 27 x

Action Output

Time Action Message

1 10:45:30 SELECT transactions.business_id FROM transactions LEFT JOIN companies ON transactions.business_id = ... 0 row(s) returned

2 10:59:25 SELECT transactions.business_id FROM transactions LEFT JOIN companies ON transactions.business_id = ... 0 row(s) returned

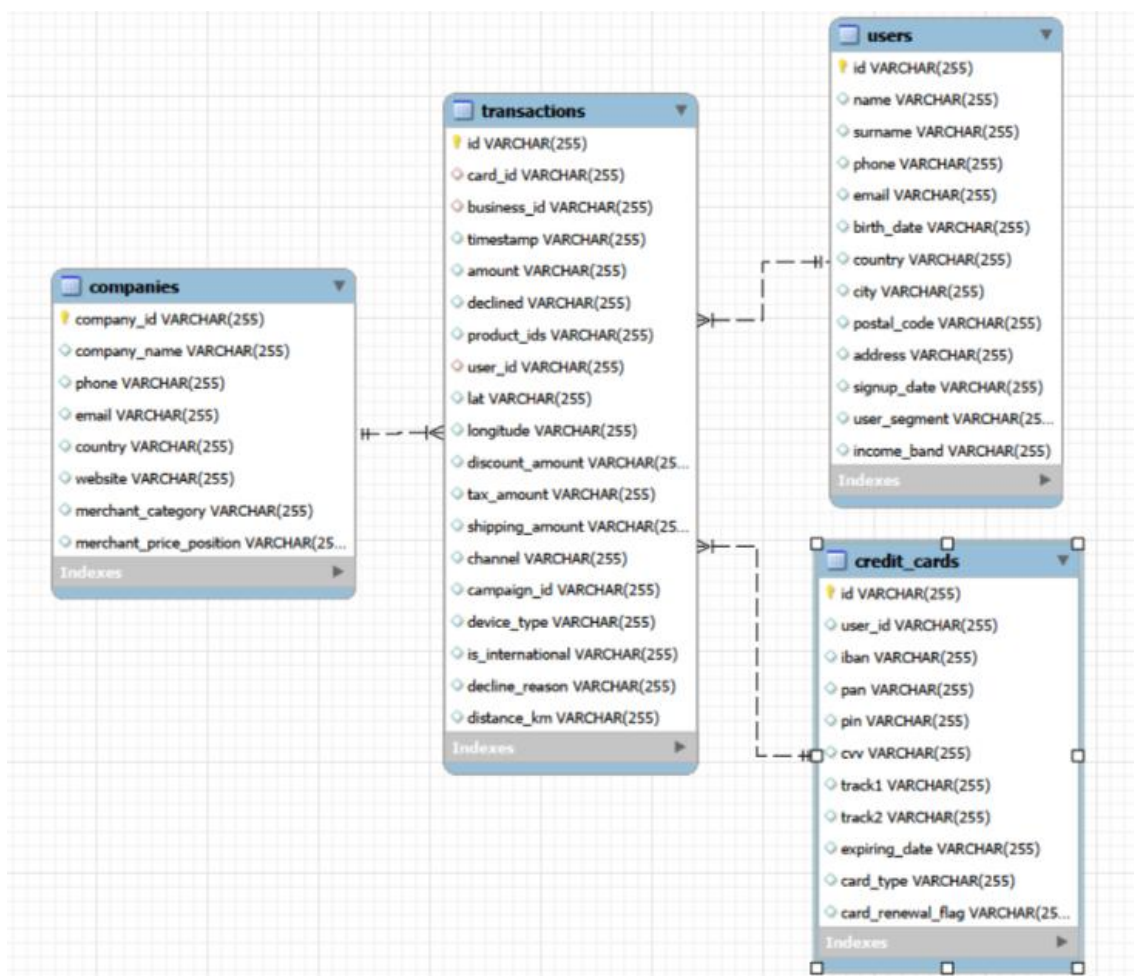
Assignar FK:

```
277 • ALTER TABLE transactions
278 ADD CONSTRAINT fk_transactions_users
279 FOREIGN KEY (user_id)
280 REFERENCES users(id);
281
282 • ALTER TABLE transactions
283 ADD CONSTRAINT fk_transactions_credit_cards
284 FOREIGN KEY (card_id)
285 REFERENCES credit_cards(id);
286
287 • ALTER TABLE transactions
288 ADD CONSTRAINT fk_transactions_companies
289 FOREIGN KEY (business_id)
290 REFERENCES companies(company_id);
```

Output

#	Time	Action	Message	Duration / Fetch
20	11:44:11	ALTER TABLE transactions ADD CONSTRAINT fk_transactions_users FOREIGN KEY (user_id) REFERENCE...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.515 sec
21	11:44:18	ALTER TABLE transactions ADD CONSTRAINT fk_transactions_credit_cards FOREIGN KEY (card_id) RE...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.500 sec
22	11:44:26	ALTER TABLE transactions ADD CONSTRAINT fk_transactions_companies FOREIGN KEY (business_id) ...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.938 sec

DIAGRAMA ESTRELLA



EXERCICI 9

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

The screenshot shows a SQL IDE interface. The top pane displays a SQL query:

```
296 SELECT *
297 FROM (
298     SELECT u.id, COUNT(t.id) AS total_transaccions
299     FROM transactions t
300     JOIN users u ON t.user_id = u.id
301     GROUP BY u.id
302     HAVING total_transaccions > 80) AS transaccions_usuari;
```

The bottom pane shows the 'Result Grid' with the following data:

id	total_transaccions
185	110
289	94
318	91
454	81

The 'Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
21	11:44:18	ALTER TABLE transactions ADD CONSTRAINT fk_transactions_credit_cards FOREIGN KEY (card_id) RE...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.500 sec
22	11:44:26	ALTER TABLE transactions ADD CONSTRAINT fk_transactions_companies FOREIGN KEY (business_id) ...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.938 sec
23	11:49:35	SELECT * FROM (SELECT u.id, COUNT(t.id) AS total_transaccions FROM transactions t JOIN users ...	4 row(s) returned	0.125 sec / 0.000 sec

EXERCICI 10

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```
304 # Exercici 10: Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules
305
306 • SELECT cc.iban, ROUND(AVG(t.amount),2) AS mitjana_amount
307 FROM transactions t
308 JOIN credit_cards cc ON t.card_id = cc.id
309 JOIN companies c ON t.business_id = c.company_id
310 WHERE c.company_name = "Donec Ltd"
311 GROUP BY cc.iban
312
```

iban	mitjana_amount
CA901355773586771626108990	361.01
US729212642816955651428791	142.56
US899450803230933276819251	246.9
US581097432668325541142297	154.01
CA441226267105857473570574	250.51

Result 70 x

Output

#	Time	Action	Message	Duration / Fetch
22	11:44:26	ALTER TABLE transactions ADD CONSTRAINT fk_transactions_companies FOREIGN KEY (business_id) ...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.938 sec
23	11:49:35	SELECT * FROM (SELECT u.id, COUNT(t.id) AS total_transaccions FROM transactions t JOIN users ...	4 row(s) returned	0.125 sec / 0.000 sec
24	11:51:59	SELECT cc.iban, ROUND(AVG(t.amount),2) AS mitjana_amount FROM transactions t JOIN credit_cards cc ...	371 row(s) returned	0.000 sec / 0.000 sec

NIVELL 2

EXERCICI 1

```
1 • USE nova_bd_transactions;
2
3 # Exercici 1: Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes.
4 # Mostra la data de cada transacció juntament amb el total de les vendes.
5
6 • SELECT DATE(timestamp) AS data_transaccio, ROUND(SUM(amount),2) AS total_vendes
7 FROM transactions t
8 WHERE "declined" = 0
9 GROUP BY DATE(timestamp)
10 ORDER BY total_vendes DESC
11 LIMIT 5;
```

data_transaccio	total_vendes
2019-11-18	15840.24
2023-02-20	15728.4
2022-12-13	15129.2
2019-03-18	14896.47
2017-12-20	14367.96

Result 33 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	12:27:53	SELECT DATE(timestamp) AS data_transaccio, ROUND(SUM(amount),2) AS total_vendes FROM transactions t WHERE "declined" = 0 GROUP BY DATE(timestamp) ORDER BY total_vendes DESC LIMIT 5;	5 row(s) returned	0.093 sec / 0.000 sec

EXERCICI 2

```
13 # Exercici 2: Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions
14 # amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024.
15 # Ordena els resultats de major a menor quantitat.
16
17 • SELECT c.company_name, c.phone, c.country, DATE(t.timestamp) AS date, ROUND(t.amount, 2) AS amount
18 FROM companies c
19 JOIN transactions t ON c.company_id = t.business_id
20 WHERE t.amount BETWEEN 350 AND 400
21 AND DATE(t.timestamp) IN ("2015-04-29", "2018-07-20", "2024-03-13")
22 ORDER BY t.amount DESC;
```

company_name	phone	country	date	amount
Fringilla LLC	+1 150 784 1716	New Zealand	2015-04-29	390.33
Auctor Mauris Vel LLP	+32 133 447 8239	United States	2024-03-13	383.58
Mauris Institute	+75 473 619 9472	Sweden	2015-04-29	369.2
Pede Cum Ltd	+26 625 798 6126	Norway	2018-07-20	358.64
Netus Et Malesuada Ltd	+52 818 402 1638	Netherlands	2024-03-13	352.21
Temper Pluen Incidunt	+89 848 141 1116	Netherlands	2024-03-13	350.71

Result 34 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	12:32:03	SELECT c.company_name, c.phone, c.country, DATE(t.timestamp) AS date, ROUND(t.amount, 2) AS amount FROM companies c JOIN transactions t ON c.company_id = t.business_id WHERE t.amount BETWEEN 350 AND 400 AND DATE(t.timestamp) IN ("2015-04-29", "2018-07-20", "2024-03-13") ORDER BY t.amount DESC;	6 row(s) returned	0.047 sec / 0.000 sec

EXERCICI 3

```
25 # Exercici 3: Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi,  
26 # per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses,  
27 # però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.  
28  
29 • SELECT business_id, COUNT(id) AS total_transaccions,  
30 CASE  
31 WHEN COUNT(id) >= 400 THEN "igual o més de 400 transaccions"  
32 ELSE "menys de 400 transaccions"  
33 END AS transaccions_llindar400  
34 FROM transactions  
35 GROUP BY business_id;  
36
```

Result Grid

business_id	total_transaccions	transaccions_llindar400
b-2222	2401	igual o més de 400 transaccions
b-2226	410	igual o més de 400 transaccions
b-2230	447	igual o més de 400 transaccions
b-2234	1514	igual o més de 400 transaccions
b-2238	472	igual o més de 400 transaccions
b-2242	449	igual o més de 400 transaccions
b-2246	1541	igual o més de 400 transaccions

Result 9

Output

#	Time	Action	Message	Duration / Fetch
1	13:37:35	SELECT business_id, COUNT(id) AS total_transaccions, CASE WHEN COUNT(id) >= 400 THEN "igual o m...	100 row(s) returned	0.234 sec / 0.000 sec

EXERCICI 4

```
37 # Exercici 4: Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.  
38  
39 • USE transactions;  
40  
41 • DELETE FROM transaction  
42 WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD";  
43
```

Output

#	Time	Action	Message	Duration / Fetch
9	13:41:51	USE transactions	0 row(s) affected	0.000 sec
10	13:41:51	select * from transaction	100001 row(s) returned	0.000 sec / 0.282 sec
11	13:46:12	DELETE FROM transaction WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD"	1 row(s) affected	0.031 sec

EXERCICI 5

```
50 • DROP VIEW IF EXISTS VistaMarketing;  
51  
52 • CREATE VIEW VistaMarketing AS  
53 SELECT c.company_name, c.phone, c.country, ROUND(AVG(amount), 2) AS mitjana_compra  
54 FROM companies c  
55 JOIN transactions t  
56 ON c.company_id = t.business_id  
57 WHERE "declined" = 0  
58 GROUP BY c.company_name, c.phone, c.country;  
59  
60 • SELECT *  
61 FROM VistaMarketing  
62 ORDER BY mitjana_compra DESC;
```

Result Grid

company_name	phone	country	mitjana_compra
Ac Fermentum Incorporated	+64 601 244 6320	Germany	330.77
Ut Semper Foundation	+10 283 500 6449	Sweden	307.37
Orci Adipiscing Limited	+60 897 390 5820	United Kingdom	306.27
Dolor Vitae Limited	+6 421 567 9195	France	305.95
Nunc In Foundation	+13 267 741 2619	Italy	305.4
Maecenas Molisusda Fnnilla Inc.	+51 671 540 5289	Netherlands	305.15

VistaMarketing 36

Output

#	Time	Action	Message	Duration / Fetch
2	12:39:08	DROP VIEW IF EXISTS VistaMarketing	0 row(s) affected	0.016 sec
3	12:39:21	CREATE VIEW VistaMarketing AS SELECT c company_name, c.phone, c.country, ROUND(AVG(amount), ...	0 row(s) affected	0.000 sec
4	12:39:40	SELECT * FROM VistaMarketing ORDER BY mitjana_compra DESC	100 row(s) returned	0.703 sec / 0.000 sec

NIVELL 3

EXERCICI 1

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en:

- 1) Si les 3 últimes transaccions han estat declinades aleshores és inactiu,
- 2) Si almenys una no és rebutjada aleshores és actiu.

Partint d'aquesta taula respon: Quantes targetes estan actives?

```
6 • USE nova_bd_transactions;
7 • DROP TABLE IF EXISTS estat_targetes;
8
9 • CREATE TABLE estat_targetes AS
10   SELECT card_id, SUM(declined) AS total_declined,
11     CASE
12       WHEN SUM(declined) >= 3 THEN 'inactiu'
13     ELSE 'actiu'
14     END AS estat_targeta
15   FROM (
16     SELECT card_id, timestamp, declined,
17       ROW_NUMBER() OVER (
18         PARTITION BY card_id
19         ORDER BY timestamp DESC
20       ) AS rn
21     FROM transactions
22   ) AS rn_table
23   WHERE rn <= 3
24   GROUP BY card_id;
```

Output

#	Time	Action	Message	Duration / Fetch
1	12:02:10	DROP TABLE IF EXISTS estat_targetes	0 row(s) affected	0.016 sec
2	12:06:01	CREATE TABLE estat_targetes AS SELECT card_id, SUM(declined) AS total_declined, CASE WHEN SUM...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0	1.078 sec

```
26 • SELECT COUNT(card_id) AS total_targetes_actives
27   FROM estat_targetes
28  WHERE estat_targeta = "actiu";
29
```

Result Grid

total_targetes_actives
5000

Result 12 x

Output

#	Time	Action	Message	Duration / Fetch
1	12:02:10	DROP TABLE IF EXISTS estat_targetes	0 row(s) affected	0.016 sec
2	12:06:01	CREATE TABLE estat_targetes AS SELECT card_id, SUM(declined) AS total_declined, CASE WHEN SUM...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0	1.078 sec
3	12:06:28	SELECT COUNT(card_id) FROM estat_targetes WHERE estat_targeta = "actiu"	1 row(s) returned	0.000 sec / 0.000 sec
4	12:06:53	SELECT COUNT(card_id) AS total_targetes_actives FROM estat_targetes WHERE estat_targeta = "actiu"	1 row(s) returned	0.000 sec / 0.000 sec
5	12:08:40	SELECT COUNT(card_id) AS total_targetes_actives FROM estat_targetes WHERE estat_targeta = "actiu"	1 row(s) returned	0.000 sec / 0.000 sec

EXERCICI 2

Crea una taula amb la qual puguem unir les dades de l'arxiu de products.csv amb la base de dades creada (ja que fins ara no podíem fer-ho):

```
34 • USE nova_bd_transactions;
35 • DROP TABLE IF EXISTS products;
36
37 • CREATE TABLE products (
38     id VARCHAR(255) PRIMARY KEY,
39     product_name varchar(255) NULL,
40     price varchar(255) NULL,
41     colour varchar(255) NULL,
42     weight varchar(255) NULL,
43     warehouse_id varchar(255) NULL,
44     category varchar(255) NULL,
45     brand varchar(255) NULL,
46     cost varchar(255) NULL,
47     launch_date varchar(255) NULL
48 );
49
50 • LOAD DATA
51 INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex_8__products.csv"
52 INTO TABLE products
53 FIELDS TERMINATED BY ','
54 ENCLOSED BY '"'
55 IGNORE 1 ROWS;
```

Output

#	Time	Action	Message	Duration / Fetch
2	12:45:13	DROP TABLE IF EXISTS products	0 row(s) affected, 1 warning(s): 1051 Unknown table 'nova_bd_transactions.products'	0.000 sec
3	12:45:22	CREATE TABLE products (id VARCHAR(255) PRIMARY KEY, product_name varchar(255) NULL, price...	0 row(s) affected	0.016 sec
4	12:45:36	LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex_8__products.csv" INTO ...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.000 sec

Tenint en compte que des de transaction tens product_ids. Genera la següent consulta:

- Necessitem conèixer el nombre de vegades que s'ha venut cada producte.

```
57 • # Comprovar si product_ids pot ser FK:
58 • SELECT t.product_ids
59 FROM transactions t
60 LEFT JOIN products p
61 ON t.product_ids = p.id
62 WHERE p.id IS NULL;
```

Result Grid

product_ids
16, 26, 97, 87
66, 69, 87
30, 11, 16, 81
35, 33, 19
93, 55, 28, 91

Result 21 x

Output

#	Time	Action	Message	Duration / Fetch
3	12:45:22	CREATE TABLE products (id VARCHAR(255) PRIMARY KEY, product_name varchar(255) NULL, ptc...	0 row(s) affected	0.016 sec
4	12:45:36	LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex_8__products.csv" INTO...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.000 sec
5	12:48:29	SELECT t.product_ids FROM transactions t LEFT JOIN products p ON t.product_ids = p.id WHERE p.id IS ...	75993 row(s) returned	0.000 sec / 0.297 sec

```
64 • # Crear tabla intermitja per poder unir FK amb PK, separant els diferents product_ids:
65
66 • DROP TABLE IF EXISTS transaction_products;
67
68 • CREATE TABLE transaction_products (
69     transaction_id VARCHAR(255),
70     product_id VARCHAR(255),
71
72     PRIMARY KEY (transaction_id, product_id)
73 );
74
75 • INSERT INTO transaction_products (transaction_id, product_id)
76 SELECT transactions.id,
77       jt_product_ids.product_id
78 FROM transactions
79 JOIN JSON_TABLE(
80     CONCAT('["', REPLACE(transactions.product_ids, ',', ', ', '"', '"'), '"']'),
81     '$[*]' COLUMNS (product_id VARCHAR(255) PATH '$')
82 ) AS jt_product_ids;
83
```

Output

#	Time	Action	Message	Duration / Fetch
6	12:50:45	DROP TABLE IF EXISTS transaction_products	0 row(s) affected, 1 warning(s): 1051 Unknown table 'nova_bd_transactions.transaction_products'	0.000 sec
7	12:51:09	CREATE TABLE transaction_products (transaction_id VARCHAR(255), product_id VARCHAR(255), ...	0 row(s) affected	0.015 sec
8	12:51:20	INSERT INTO transaction_products (transaction_id, product_id) SELECT transactions.id, jt_product_id...	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	6.375 sec

```
87 • ALTER TABLE transaction_products
88 ADD CONSTRAINT fk_transaction_id
89 FOREIGN KEY (transaction_id)
90 REFERENCES transactions(id);
91
92 • ALTER TABLE transaction_products
93 ADD CONSTRAINT fk_product_id
94 FOREIGN KEY (product_id)
95 REFERENCES products(id);
96
97 • SELECT product_id, COUNT(transaction_id) AS total_vendes
98 FROM transaction_products
99 GROUP BY product_id
```

product_id	total_vendes
1	2468
10	2571
100	2517
11	2573
12	2558

Result 23 x

Output

#	Time	Action	Message	Duration / Fetch
1	12:53:58	ALTER TABLE transaction_products ADD CONSTRAINT fk_transaction_id FOREIGN KEY (transaction_id) ...	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	2.953 sec
2	12:54:07	ALTER TABLE transaction_products ADD CONSTRAINT fk_product_id FOREIGN KEY (product_id) REFER...	253391 row(s) affected Records: 253391 Duplicates: 0 Warnings: 0	3.906 sec
3	12:54:17	SELECT product_id, COUNT(transaction_id) AS total_vendes FROM transaction_products GROUP BY produ...	100 row(s) returned	0.062 sec / 0.000 sec